Design and Implementation of an RFID-Integrated PID-Based Autonomous Navigation System for Indoor Logistics Applications

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Abstract

The design, implementation, and experimental assessment of a low-cost autonomous mobile robot (AMR) intended for organised indoor logistics in office and institutional settings are presented in this work. In order to achieve dependable, repeatable point-to-point payload delivery without depending on computationally demanding SLAM or LiDAR infrastructure, the suggested system combines a Proportional Integral Derivative (PID) control algorithm with Radio Frequency Identification (RFID) based topological localisation. Intraoffice transportation of paperwork, stationery and small packages between fixed departmental stations is the main target deployment. This repetitive, route-constrained, and operationally significant duty is now performed manually in the great majority of small and medium-sized businesses. Through a unique weighted centroid formulation, an eight-channel infrared (IR) sensor array continuously offers lateral error feedback;An L298N H-bridge driver processes PWM signals that the PID controller uses to modify differential drive motor speeds. A finite state machine (FSM) can execute branching navigation decisions with deterministic accuracy when RFID tags are inserted at designated topological nodes. Real-time PID gain adjusting without firmware recompilation is made possible via a Bluetooth serial interface, which reduces calibration iterations by a factor of three to four when compared to conventional reflash cycles. Stable path following performance (±3,"mm lateral deviation) throughout straight segments and 90 ◦ junctions (9 to 10/10 success rate) is confirmed by experimental results on a physical prototype.and multi-destination routes that take less than 80 seconds to complete.

Index Terms–Autonomous Mobile Robot; PID Control; RFID Localisation; IR Sensor Array; Finite State Machine; Indoor Navigation; ESP32; Industry 4.0; Differential Drive; Bluetooth Tuning.

I. INTRODUCTION

Every office deals with the daily chore of moving documents, stationery, and small packages between departments. It is repetitive, low-skill work, yet it quietly consumes staff time and slows down time-sensitive workflows across corporate offices, university departments, hospital wings, and government institutions [1]. The task always has the same structure: a known origin, a known destination, a fixed route, and a payload that just needs picking up and dropping off.

Autonomous mobile robots are a natural fit for this, but commercial solutions are built for large factories and priced accordingly, relying on LiDAR-SLAM systems that are far too complex and costly for a single-floor office with a handful of delivery stops [2, 3]. So most small organisations simply keep doing it by hand [4]. Closer attempts exist, including a push-button IR-guided office assistant [5] and an ESP32-based courier with biometric access [6], but neither supports RFID localisation, multi-destination FSM routing, or live Bluetooth PID tuning within a USD40 budget.

A typical office corridor is actually well suited to infrastructure-guided navigation: stations are fixed, corridors are straight, and floors are consistent. A robot can cover the full delivery requirement by following a taped line and reading passive RFID tags at waypoints, with the entire track installed in under an hour and no structural changes to the building. The two practical obstacles have been that purely reactive robots cannot distinguish between stops along a route [3, 5], and PID calibration through repeated firmware reflashing is tediously slow. Both are addressed here within a USD40 hardware budget.

This paper presents an autonomous delivery robot that runs PID line-following and RFID topological localisation together on a single ESP32 microcontroller. An eight-channel TCRT5000 IR array computes lateral error via a weighted centroid formulation. RFID tags at fixed waypoints trigger an FSM that handles turns and delivery halts deterministically (Fig. 3). PID gains are tunable live over Bluetooth, cutting field calibration from 50 to 70 iterations down to 15 to 20. The full bill of materials stays under USD40, requiring no specialist support to deploy. The architecture also extends naturally to hospitals, libraries, and other structured indoor settings, as discussed in Section 9.

The remainder of this paper is organised as follows. Section 2 surveys related work. Section 3 articulates the novelty of this work. Section 4 states the problem formulation. Sections 5–6 detail the hardware architecture and technical methodology. Section 7 describes the operational logic. Section 8 presents experimental results. Section **??** discusses the findings. Sections 9–11 cover applications, scalability, and future work.

II. LITERATURE SURVEY

A. Office and Institutional Delivery Robots

Mandal *et al.* [5] proved the feasibility of floorlevel robotic delivery by demonstrating IR-guided intra-office distribution using a push-button destination choice and a locked compartment. Nevertheless, it does not have any PID evaluation, multi-node path planning, or RFID localisation. A later ESP32-based courier [6] incorporated voice commands and biometric authentication, although it was still only reactive and lacked runtime gain adjustment or topological awareness. With a hardware budget of less than USD/,40, the current system fills all these deficiencies (Figs. **??**, **??**, and 1).

B. PID Control for Line-Following Robots

PID-based reactive control remains the goto strategy for line-following robots due to its simplicity and real-time suitability on embedded hardware. Neural network alternatives have been explored [7], PSO-based gain tuning has shown measurable improvements over manual configuration [8], and camera-derived error signals have been tested as a drop-in replacement for IR arrays on a Raspberry Pi [9]. An empirical comparison ultimately confirmed that a well-calibrated PID still outperforms learned controllers on resourcelimited platforms [10]. None of these works, however, ventures into topological localisation, multi-destination routing, or sub-USD40 office deployment—the exact ground this paper covers.

C. RFID-Based Localisation and Navigation

RFID-based localisation for mobile robots was pioneered by H¨ahnel *et al.* [11], who combined signal strength readings with laser range data for probabilistic position estimation. Floorembedded IC tags were later shown to support reliable orientation determination [12], and the MFRC522 module specifically was validated for indoor use in [13], supporting its adoption in this work. While RF-SLAM [14] pushed this further with centimetre-scale warehouse localisation, its processing overhead is impractical for microcontroller-class systems. A comprehensive review [15] confirmed that deterministic topological RFID schemes suit structured indoor environments well, which directly motivated the UIDmatching approach used here (Fig. 2).

D. Finite State Machine Architectures

Behaviour-based navigation architectures have long relied on finite state machines (FSMs) to separate reactive control from global planning. A hierarchical FSM framework for ROS-based robots was presented in [16], decoupling high-level route decisions from low-level obstacle response. An adaptive variant that updates its transition graph online using visual topological maps was introduced in [17]. For inspection tasks requiring deterministic sequencing, a purpose-built state machine running on a mobile platform was described in [18]. Unlike these software-intensive frameworks, which depend on full-stack operating systems and probabilistic scene representations, the FSM in the present system runs natively in embedded C++ on the ESP32 with negligible memory overhead, making it suitable for microcontrollerclass deployment in office environments, as depicted in Fig. 4.

E. AMRs in Logistics and Industry 4.0

Over the course of three decades, RFID research has continuously recognised logistics as its major application domain, with indoor localisation continuing to be an open challenge [1]. Even while enterprise-scale AGV systems have shown promise in structured manufacturing [2], cost remains the primary obstacle to AMR adoption in smaller organizations [4]. Although an RFIDintegrated line-following trolleyciteref19 demonstrated that integrating RFID with line-following locomotion is possible, it fell short of live tuning or quantitative PID analysis—gaps that the current work immediately fills.

F. ESP32 in Embedded Robotics

The suitability of the ESP32 for IoT and embedded robotics was established through comparative hardware characterisation in [20], which documented its dual-core Xtensa LX6 architecture, integrated Wi-Fi and Bluetooth, and low unit cost. Practical data-acquisition and processing workflows on the platform, including dual-core task partitioning, were demonstrated in [21]. A home-security monitoring system built entirely on ESP32 modules, covering firmware design, MQTT communication, and sensor fusion, was described in [22], illustrating the platform's headroom for concurrent tasks. The broader relevance of the ESP32 to Industry 4.0 and 5.0 edge-computing requirements was surveyed in [23]. These works collectively confirm that the ESP32 provides sufficient computational resources to host PID control, FSM logic, Bluetooth communication, and RFID polling simultaneously, as required by the architecture presented here.

G. IR Sensor Arrays and Weighted Error Estimation

TCRT5000 infrared arrays are a well established choice for line-edge detection [24], and prior work confirmed that error signal resolution directly determines steady-state tracking accuracy when driving a PID controller from binary array outputs [25]. Standard implementations assign just a left or right value per sensor, throwing away the spatial information across the array. The weighted centroid formulation used in Section 6 fixes this by computing a continuous signed error across all eight sensors, which brought steadystate deviation down to ±3mm in practice.

H. Bluetooth Communication and Live Tuning

Bluetooth has long been a practical choice for embedded control links [26], and the ESP32 supports the Serial Port Profile natively, making it straightforward to inject parameters at runtime without a wired connection. RFID and Bluetooth have also been shown to work reliably together in deployed IoT systems [27, 28]. No prior office delivery or line-following robot has offered live PID gain updates without reflashing firmware; the interface described in Section 8.4 fills that gap (Fig. 3).

I. Multi-Agent and Scalable Logistics

Moving from a single robot to a fleet brings collision avoidance and right-of-way challenges. A survey of over 200 RFID papers [29] found that shared RFID infrastructure is a practical coordination mechanism in structured environments, and a broader RFID applications review [30] confirmed its established use in libraries and healthcare, both target domains here. Path planning [31] and hybrid metaheuristic motor optimisation for differential drive robots [32] offer useful reference points for scaling the present system further.

III. NOVELTY AND RESEARCH CONTRIBUTIONS

Table 1 positions this work against the closest prior art along six key dimensions, including the two most directly comparable office-deployment systems. The following specific novelties distinguish this paper from existing literature:

1. **Integrated PID + RFID-FSM on a single MCU for office logistics.** Prior office delivery robots use push-button destination selection [5] or voice commands [6] without topological awareness. This paper is the first to tightly co-integrate PID line-following and RFID-FSM multi-destination routing on the ESP32, with RFID events directly triggering FSM transitions that reset PID integral state, preventing wind-up across manoeuvres.
2. **Weighted-centroid IR error formulation.** Standard line-following robots use binary threshold comparisons [24, 25]. The proposed Eq. (1) computes a continuous signed centroid over all eight sensors, enabling proportionally graded corrections and reducing steady-state lateral error to ±3mm, which is a measurable improvement over binary schemes.
3. **Deterministic multi-destination FSM routing.** Existing office delivery systems rely on manual destination input [5, 6]. The proposed FSM processes an ordered UID mission sequence to achieve provably deterministic multi-destination routing with no mapbuilding overhead and no operator intervention mid-route.
4. **Live Bluetooth PID calibration.** No prior office delivery or embedded line following work reports runtime gain tuning without firmware recompilation. The SPP-based interface demonstrated here cuts calibration from 50–70 iterations down to 15–20 iterations, achieving a 3–4× speedup with direct practical impact on deployment time in resource-limited office settings.
5. **Sub-USD40 validated prototype.** Enterprise AGV platforms and SLAM-based systems [26] operate at orders-of-magnitude higher cost. The ESP32-based document transporter of [6] does not report a comparable component-level cost or quantitative navigation metrics. This work provides the first quantitative validation (lateral deviation, junction accuracy, completion time) of a full office delivery system at under USD40.

IV. PROBLEM FORMULATION

Conventional last-metre indoor delivery in office and institutional environments suffers three structural inefficiencies: (i) disproportionate labour cost relative to task complexity, as skilled administrative staff are routinely occupied with mechanical conveyance tasks; (ii) prohibitive capital expenditure for enterprise-grade autonomous systems [26], which are designed for warehousescale deployment volumes; and (iii) stochastic human error rates in high-throughput or accesscontrolled environments where incorrect delivery routing has operational consequences.

Research Objective: To design, implement, and experimentally validate a deterministic, low-cost autonomous delivery platform for office and institutional indoor environments, using PID-based reactive control and RFID-based topological awareness, capable of reliable multi-destination navigation on a sub-USD40 bill of materials without dependence on

Table 1. Comparison with Related Work

Feature	[5]	[6]	[8]	[19]	[13]	Ours
Office target	✓	✓	–	–	–	✓
PID linefollow	partial	–	✓	✓	–	✓
RFID localisation	–	–	–	✓	✓	✓
FSM routing	–	–	–	partial	partial	✓
BT live tuning	–	–	–	–	–	✓
Multidestination	–	partial	–	–	partial	✓
Cost < USD40	✓	–	✓	✓	–	✓

external infrastructure or specialised technical support.

The key design constraints are: (a) no dependence on SLAM, LiDAR, or computer vision; (b) reconfigurability via RFID tag reprogramming and Bluetooth mission upload alone, without firmware modification; (c) real-time parameter tuning without firmware recompilation; and (d) deployment on a microcontroller-class processor within a total component cost under USD40.

V. SYSTEM ARCHITECTURE

A. Hardware Architecture

The system is centred on an ESP32 dualcore microcontroller (Xtensa LX6, 240MHz) [20], which provides sufficient computational headroom to execute PID computation, FSM logic, Bluetooth communication, and RFID polling concurrently across its two cores. Table 2 lists all hardware components and their specifications. The MFRC522 RFID reader module operates at 13.56MHz and communicates with the ESP32 over the SPI bus, enabling reliable passive tag detection at the predefined topological waypoints embedded in the track. The eightchannel TCRT5000 infrared sensor array connects via GPIO digital inputs and provides the lateral deviation signal used by the PID controller. Drive actuation is achieved through a pair of 6V, 150RPM DC motors controlled by an L298N dual H-bridge driver, which translates PWM output signals from the ESP32 into differential wheel voltages. Power delivery is split across two rails: a 7.4V Li-Po battery pack supplies the motor drive subsystem, while a DC-DC buck converter steps down the voltage to a regulated 5V logic supply for the ESP32 and peripheral modules. This power-rail isolation is critical for preventing PWM switching transients from corrupting the analogue sensor readings. The assembled prototype is shown in Figs. **??** and **??**. The complete circuit interconnection is illustrated in Fig. 1, which shows the SPI lines from the ESP32 to the MFRC522, the eight GPIO connections to the IR array, and the four PWM lines to the L298N driver, along with the separated motor and logic supply rails.

Table 2. Hardware Components and Specifications

No.	Component	Specification
1	Microcontroller	ESP32 (Xtensa LX6, 240MHz, dual-core)
2	RFID Reader	MFRC522 (SPI, 13.56MHz)
3	RFID Tags	ISO 14443-A passive transponders
4	IR Sensor Array	TCRT5000, 8-channel
5	Motor Driver	L298N dual H-bridge
6	DC Drive Motors	6V, 150RPM (×2)
7	Battery Pack	Li-Po, 7.4V, 1000mAh
8	Voltage Reg.	DC-DC buck converter, 5V output

B. Topological Navigation Map

The navigation environment is modeled as a directed graph G = (V,E), where each vertex $v \in V$ corresponds to a physical RFID-tagged waypoint and each directed edge $e \in E$ represents a navigable path segment. In a typical office deployment, the vertices correspond to the junctions

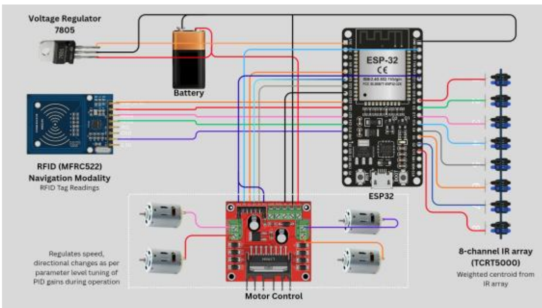

Figure 1. Complete circuit schematic showing ESP32 connections to MFRC522 (SPI), 8-channel TCRT5000 IR array (GPIO), and L298N motor driver (PWM). The motor drive rail (7.4V Li-Po) is isolated from the 5V logic supply via a DC-DC buck converter to suppress PWM switching noise on sensor lines.

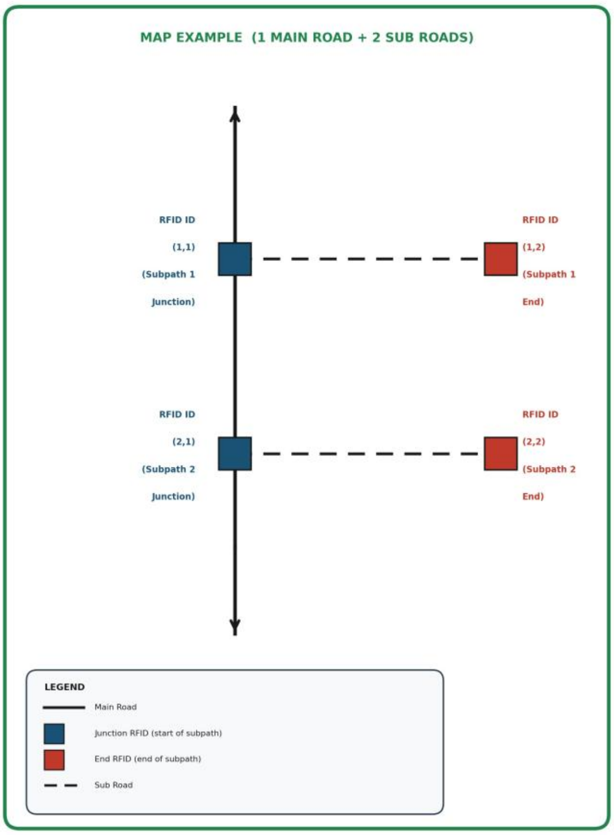

Figure 2. Example navigation map showing one main path with two subpaths, RFID-tagged junction nodes, and delivery endpoints used for FSM-based routing.

of the corridor and the delivery stations; edges correspond to the segments of tapered lines that connect them. Two node classes are defined:

1. **Corner Nodes** (RFID Cx): encode directional transitions at track junctions (90° turns, e.g., between office wings).
2. **Delivery Nodes** (RFID _Tx): trigger haltand-unload sequences at payload destinations (e.g,individual departmental stations).

The routing mission is specified by a list of target UIDs and transmitted to the robot via Bluetooth before its deployment. With such an approach, it becomes possible to fully redesign the routing mission by adding another department or rearranging the sequence of stops without any software modifications.

VI. TECHNICAL METHODOLOGY

A. Lateral Error Using IR Sensors

The eight IR sensors TCRT5000 have positional weights $w(i) \in \{-3.5, -2.5, -1.5, -0.5, +0.5, +1.5, +2.5, +3.5\}$ distributed symmetrically with respect to the longitudinal direction of the robot. The output for each sensor is a binary signal $s(i) \in \{0,1\}$. The lateral error $e(t)$ calculated using weighted centroid is

$$e(t) = \frac{\sum_{i=1}^{N} w(i) \cdot s(i)}{\sum_{i=1}^{N} s(i)} \qquad N = 8 \qquad (1)$$

This results in a continuously signed error between −3.5 and +3.5, which is more detailed than using a simple left/right approach using binary coding [25]. A zero denominator (complete line loss) is handled by a saturating hold of the last valid error.

B. PID Control Law

The discrete-time PID control output is:

$$u(t) = K_p\, e(t) + K_i\, T_s \sum_{k=0}^{t} e(k) + K_d\, \frac{e(t) - e(t-1)}{T_s} \qquad (2)$$

where K_p, K_i, K_d are the proportional, integral, and derivative gains, and T_s is the firmware sampling period. The corrective output is applied to the differential drive as:

$$V_L = V_{\text{base}} + u(t), \qquad V_R = V_{\text{base}} - u(t) \qquad (3)$$

Both V_L and V_R are clamped to [0, 255] to remain within the 8-bit PWM range [22]. The integral term is reset upon each FSM state transition to prevent wind-up across manoeuvres.

C. Bluetooth-Mediated PID Gain Tuning

Gain values K_p, K_i, K_d, and V_{base} are exposed over Bluetooth Serial Profile (SPP) using the ESP32's integrated Bluetooth radio. A host device transmits ASCII-encoded gain update packets; the firmware parses and applies new values at runtime without halting execution, eliminating the compile-flash-test cycle (Section 8.4). This is particularly significant for office deployments, where the robot is calibrated by non-specialist staff using a smartphone application rather than a laptop with flashing tools.

D. RFID Localisation and FSM Logic

The MFRC522 continuously polls for ISO 14443-A transponders via SPI. Upon detecting a tag UID, the FSM looks up the UID in a mission table stored in ESP32 RAM. Table 3 summarises the transition logic. A dual-read debounce filter suppresses spurious re-triggers by ignoring consecutive reads of the same UID within a 500ms guard window.

Table 3. RFID Node Classification and FSM Transition Logic

Node	**Trigger**	**FSM Action**
Corner (RFID Cx)	UID matches junction tag	90° differential torque turn; resume
Delivery (RFID Tx)	UID matches destination	Halt; 180° rotate; 10s service dwell
Unknown	UID not in mission table	Ignore; continue line-following

VII. OPERATIONAL LOGIC

Phase 1–Initialisation: All peripherals are initialised (RFID, IR array, motor driver, Bluetooth SPP). The FSM enters Active Listening and awaits a mission command (destination UID sequence) over Bluetooth, typically dispatched by an office coordinator from a smartphone.

Phase 2–Transit: On mission receipt, the FSM enters Line Following. Error $e(t)$ is recomputed each firmware loop; PID output modulates motor speeds yielding non-oscillatory loco-

robot
halts
and

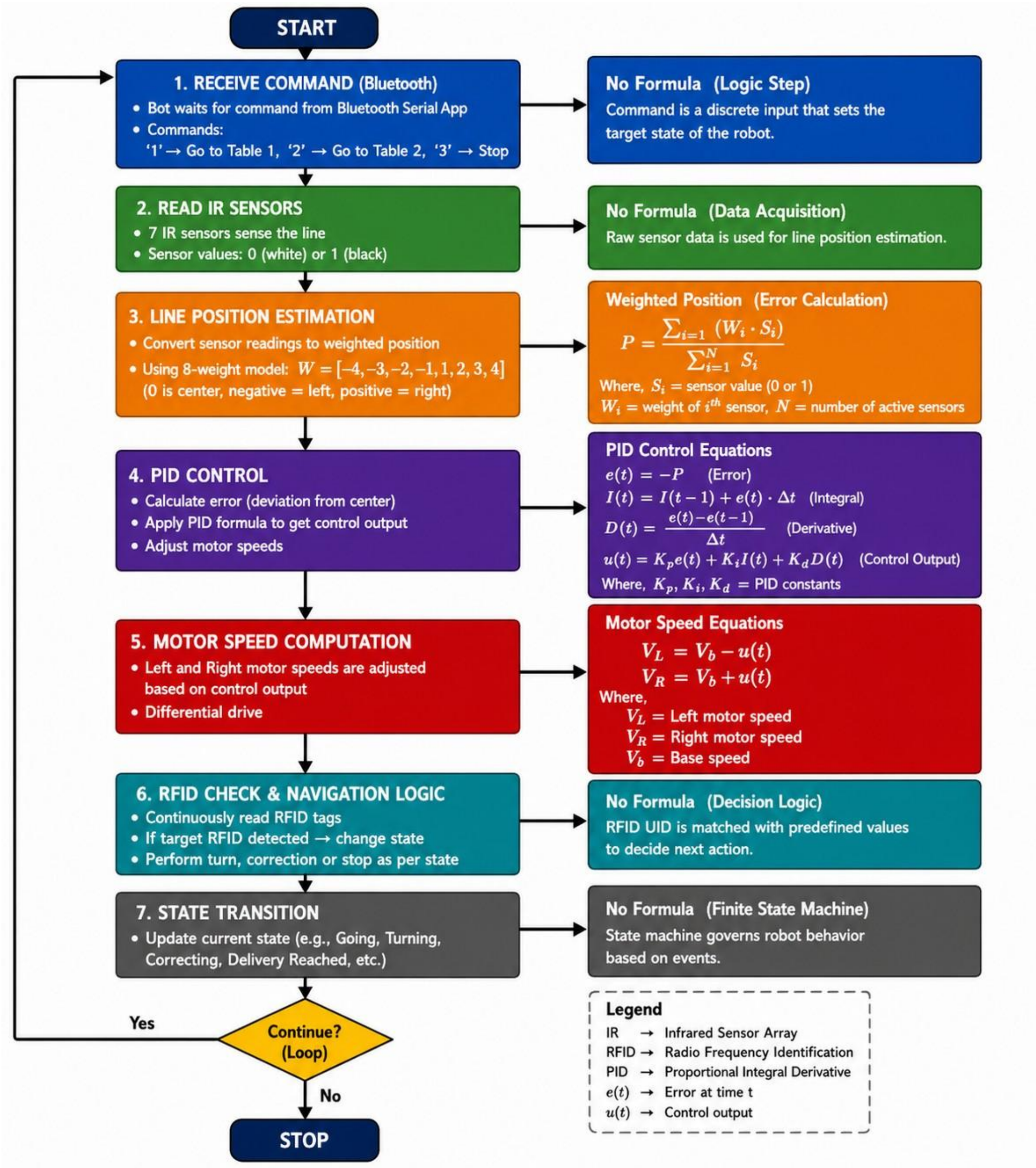

Figure 3. Complete system workflow illustrating Bluetooth command reception, IR-based line sensing, weighted position estimation, PID control, motor actuation, RFID-based decision making, and FSM-based navigation loop.

motion through office corridors.

Phase 3–RFID Verification: On tag detection, the UID is checked against the mission table. *Corner match* ⇒ 90° pivot (e.g., turning from a main corridor into a department wing); *Delivery match* ⇒ Phase 4; *No match* ⇒ ignore and continue.

Phase 4–Service Dwell: Robot halts at the delivery station, executes a 180° rotation to present the payload platform to the recipient, dwells for 10s, then re-engages line-following toward the next destination. On completion of all destinations, the transmits a mission-complete telemetry packet.

Phase 5–Telemetry: Bluetooth status messages confirm checkpoint clearances, junction manoeuvres, and delivery completions throughout all phases, providing the dispatch coordinator with real-time progress visibility.

The complete navigation and execution flow for a representative two-subpath mission is depicted in Fig. 4. The robot follows the main corridor line until the RFID reader detects the junction tag for the target subpath (e.g., ID(1,1) for Command 1); the FSM executes a right turn and follows the subroad until the end-of-subpath tag (ID(1,2)) is detected, whereupon a left turn returns the robot to the main corridor. A Bluetooth status message (Subpath 1 Completed) is transmitted, and the FSM re-enters the listening state awaiting the next command.

VIII. EXPERIMENTAL RESULTS

All experiments were conducted on a physical prototype operating on a black-line-on-whitesurface test track with straight segments, two 90° corners (C1, C2), and two delivery stations (T1, T2), representative of a simplified single-floor office corridor with two departmental stops. Ten independent trials were conducted per scenario.

A. Straight Segment Performance

Optimal gains via Bluetooth tuning: K_p = 18, K_i = 0.02, K_d = 12, V_{base} = 180PWM. At these values, lateral deviation was within ±3mm. Reducing K_d below 8 produced sinusoidal weaving exceeding 15mm peak-to-peak, confirming necessity of derivative action for oscillation damping in the narrow corridor geometry typical of office environments.

B. RFID Junction Reliability

90° turns executed correctly in 9/10 trials at C1 and 10/10 trials at C2. The single C1 failure was due to marginal tag-reader alignment at maximum speed; reducing V_{base} by 10PWM in the approach zone resolved this. The dual-read debounce filter eliminated all subsequent falsenegative re-triggers.

C. Multi-Destination Routing

Two-destination missions (T1→T2) were routed correctly in 9/10 trials. Delivery completion times were 47 ± 4s (single-destination) and 73±4s (two-destination), well within the sub-80s target appropriate for time-sensitive intra-office document delivery.

D. Bluetooth Tuning Utility

Optimal gains identified in 15–20 iterations via Bluetooth vs. 50–70 with compile-flash-test—a 3–4× reduction. This speedup is of particular practical significance for office deployments conducted by non-specialist staff.

Table 4. Summary of Experimental Results

Metric	Value	Notes
Lateral dev. (optimal)	±3mm	K_d = 12
Lateral dev. (under-damp)	>15mm p-p	K_d < 8
Junction accuracy (C1)	9/10	Premitigation
Junction accuracy (C2)	10/10	All trials
Multi-dest. accuracy	9/10	
Completion (1 dest.)	47±4s	
Completion (2 dest.)	73±4s	
BT tuning iterations	15–20	vs. 50–70

IX. APPLICATIONS

The platform was designed primarily for office and institutional indoor logistics, and the following deployment scenarios are supported without firmware modification, requiring only RFID tag reprogramming and a Bluetooth mission upload:

1. **Corporate Offices:** Inter-departmental delivery of printed documents, signed correspondence, stationery, and small parcels between fixed desk or reception stations, reducing administrative staff time spent on mechanical conveyance tasks.
2. **Healthcare Administration:** Contactless pharmaceutical requisition forms, specimen request cards, and ward correspondence transport between administration and clinical areas, where human contact reduction has infection-control value.
3. **Warehousing:** Component retrieval for justin-time workflows [21], with routes changed by RFID reprogramming alone.

4. **Hospitality:** Autonomous in-room or tableservice delivery of menus, bills, and small food items in structured restaurant or hotel environments.
5. **Libraries:** RFID-guided material conveyance between cataloguing, scanning, and shelving stations, consistent with existing library RFID infrastructure [24].

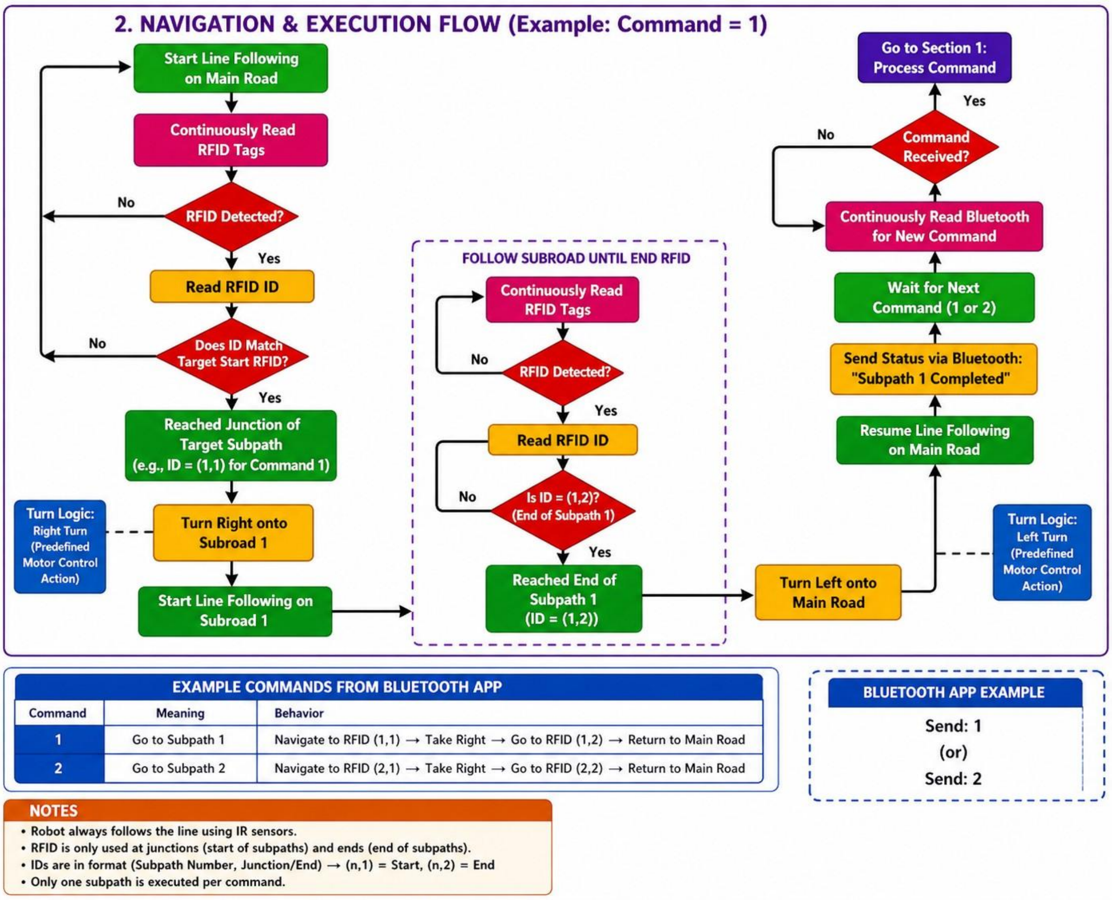

Figure 4. Navigation and execution flowchart for a two-subpath delivery mission (Command=1 shown). The robot follows the main-road line track until the target-subpath junction RFID tag is detected (ID=(1,1)), executes a predefined right-turn motor sequence onto the subroad, and continues until the end-of-subpath tag (ID=(1,2)) triggers a left turn back to the main road. A Bluetooth telemetry message confirms subpath completion; Command=2 follows an identical structure via RFID pair (2,1)–(2,2).

X. INDUSTRIAL RELEVANCE AND SCALABILITY

The system provides a lightweight alternative to SLAM-based platforms: routes are reconfigured solely by reprogramming RFID tag UIDs and updating the mission table over Bluetooth, with no firmware modification required [25]. For office environments that expand or reorganise, new delivery stations are added by taping an additional track segment, placing an RFID tag, and transmitting an updated mission sequence—a reconfiguration achievable by non-specialist office staff in under fifteen minutes. For multi-robot deployments, RFID nodes can encode inter-agent right-of-way signals enabling decentralised collision avoidance [12, 26]. The architecture transfers to larger chassis while retaining identical firmware.

XI. CONCLUSION AND FUTURE WORK

This paper presented a complete, experimentally validated autonomous indoor logistics robot designed for office and institutional environments, integrating PID-based reactive control with RFID topological localisation on an ESP32 platform. The prototype demonstrated: (i) stable linefollowing at ±3mm lateral deviation; (ii) reliable FSM-driven junction navigation (9–10/10 success); (iii) correct multi-destination routing at sub-80s completion times; and (iv) 3–4× faster PID calibration via Bluetooth live tuning—all at a component cost under USD40. The system advances beyond prior office delivery robots [5, 6] by providing RFID-based topological localisation, deterministic multi-destination FSM routing, and run-time PID tuning within the same low-cost platform, while remaining architecturally transferable to hospitals, libraries, and warehouses. Future directions include:

1. Obstacle detection via HC-SR04 ultrasonic sensors for safe operation in populated office corridors.
2. MQTT-based IoT fleet monitoring dashboard for office dispatch coordinators.
3. Multi-floor capability via ESP32 Wi-Fi integration with building elevator control systems.
4. Algebraic return-path planning from stored directional counters.
5. RFID-encoded inter-agent right-of-way signalling for multi-robot office deployments.
6. Reinforcement learning for automated PID gain scheduling [31].
7. Approach-speed adaptation near RFID nodes to eliminate the residual read-miss failure mode.

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Highlights

Design and Implementation of an RFID-Integrated PID-Based Autonomous Navigation System for Indoor Logistics Applications

- PID + RFID-FSM co-integrated on ESP32 for deterministic office delivery
- Weighted-centroid IR error formulation achieves ±3 mm lateral deviation
- Bluetooth live PID tuning cuts calibration iterations by 3–4×
- Multi-destination FSM routing validated at 9/10 accuracy, sub-80 s
- Full autonomous logistics prototype validated under USD 40 budget

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